# CAREER: Foundations of Infrastructure-Indexed Energy for Scalable AI and Computing

Yi Ding, yiding@purdue.edu

## Overview

The rapid growth of artificial intelligence (AI) and large scale computing systems is driving unprecedented demand for energy and power. Improving energy efficiency is therefore a central challenge in computing systems. However, existing approaches assume that energy is uniform, meaning that a unit of energy has the same system level cost regardless of where and when it is consumed. In practice, the physical systems that produce and deliver energy, including power generation, electric grids, and datacenter facilities, vary across location and time, leading to differences in price, capacity constraints, and pollutant intensity.

This project introduces *Infrastructure Indexed Energy (IEI)*, a new systems abstraction that models energy as a non uniform, location and time dependent quantity. Infrastructure is treated as the physical systems that produce and deliver energy for computing, and energy cost is defined as a system level function that maps energy use to infrastructure dependent consequences. The project will develop generalizable models and methods that connect system decisions across hardware, datacenters, and AI workloads to infrastructure dependent energy cost signals, enabling systems to optimize how, where, and when energy is used.

**Keywords:** Energy-efficient computing; AI infrastructure; lifecycle modeling; resource-aware optimization

## Intellectual Merit

This project addresses a fundamental gap in computing systems: the lack of abstractions and models that capture non uniform energy cost across infrastructure. The central hypothesis is that modeling energy as an infrastructure indexed quantity enables new classes of system level optimizations that are not achievable under uniform energy assumptions.

The research will (1) develop lifecycle aware models that map computing activity to infrastructure dependent energy cost signals, (2) construct fine-grained, spatial and temporal signals that expose variation across deployment conditions, and (3) design algorithms and optimization frameworks that incorporate these signals into system design and runtime decision making. The project will establish IEI as a general abstraction for reasoning about energy in computing systems and demonstrate its impact across multiple system layers. It will analyze tradeoffs between efficiency, performance, and infrastructure dependent cost signals, providing new insights into system design under realistic constraints. The resulting methods will be evaluated across representative workloads and platforms to demonstrate generality and practical applicability.

## Broader Impacts

This project will advance U.S. leadership in computing by enabling more efficient and robust AI and computing infrastructure. The proposed models and methods can improve energy utilization, support scalable system deployment, and inform infrastructure level decision making.

The project will contribute to workforce development by training students in systems, AI, and resource constrained computing, and will release open source models and tools to accelerate research and adoption. The results will provide actionable insights for industry and public sector stakeholders involved in the design and operation of computing infrastructure in the United States. It will also strengthen collaboration between academia and industry by aligning system design with real world infrastructure challenges faced by U.S. organizations. Outreach and educational activities will broaden participation and prepare a diverse pipeline of students for careers in computing systems and AI.